Polynomial identities

1

a $a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2$

b $a^2c^2 + b^2d^2 + b^2c^2 + a^2d^2$

2 No solution provided.

3 $1\,004\,004$

4 No solution provided.

5 4080

6 15689

7

a $(u+v)(u-v)$ **b** 5400 **c** 384

8 No solution provided.

9 No solution provided.

10 $9\,010\,000$

Applications

11

a 136

b Example answer:
$x^2 \times y^2 = x^2y^2$
$= (xy)^2$

12

a 16 **b** 49 **c** t^2 **d** 4900

e No

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13

a $10m + n$

b $10n + m$

c $(10m + n) - (10n + m) = 9(m - n)$

14

a $15, 16, 17$

b **C** and **D**

15

a $m^4 + 2m^2n^2 + n^4$

b No solution provided.

c Pythagora's Theorem

d $16, 34$

16 $100, 28, 96$

17

a $x^3 - a^3$

b $97 \times 10\,309$

18

a $2x^4 + 5x^3 + 8x^2 + 6x + 4$

b 25864

c 10

19

a $x^2 = (x - 1)^2 + x - 1 + x$

b No solution provided.

c 10201

20

a 121

b 11

c
$$\begin{aligned} x(x+1)(x+2)(x+3)+1 &= \left(x^2+x\right)\left(x^2+5x+6\right)+1 \\ &= x^4+6x^3+11x^2+6x+1 \\ &= \left(x^2+3x+1\right)\left(x^2+3x+1\right) \\ &= \left(x^2+3x+1\right)^2 \end{aligned}$$

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